# AI for Office & Productivity

A hands-on, four-hour deep dive into transforming how you work — from emails to spreadsheets to dashboards — using Google Gemini, ChatGPT, and the AI tools built right into your everyday apps.

Designed for practical adoption, this workshop focuses on everyday tasks you can automate, speed up, and improve immediately.

01

## Hour 1

**AI fundamentals for office work**

- What Gemini and ChatGPT are best at
- Writing better prompts for common office tasks
- Turning rough notes into polished emails, summaries, and plans

**Example:** Draft a client update, rewrite it in a more professional tone, then shorten it for Slack.

02

## Hour 2

**Spreadsheet automation with AI**

- Clean and categorize messy data faster
- Create formulas, summaries, and conditional logic with AI assistance
- Build faster workflows in Excel and Google Sheets

**Example:** Ask AI to generate a formula for tracking overdue invoices or identifying duplicate records.

03

## Hour 3

**Data visualization & insights**

- Choose the right chart for your message
- Turn tables into clear charts and dashboards
- Use AI to summarize trends, outliers, and next steps

**Example:** Convert a monthly sales table into a chart and ask AI to highlight the biggest change drivers.

04

## Hour 4

**Hands-on practice & real-world applications**

- Work through realistic office scenarios
- Practice prompting, reviewing, and refining outputs
- Apply AI to your own workflows and tools

**Example:** Rework a meeting summary into action items, assign owners, and create a follow-up email.

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# Practice resources

Use free public datasets to test AI prompts, spreadsheet cleanup, and dashboard building during the workshop:

- **Kaggle Datasets**
- **Google Dataset Search**
- **data.world free datasets**

HOUR 1

# AI for Office & Productivity

AI can save time on everyday office work by drafting content, summarizing information, analyzing data, and automating repetitive tasks. In this first hour, we focus on the basics of generative AI, how models are trained, and where they fit into practical workplace workflows.

## What is Generative AI?

Generative AI refers to systems trained on large amounts of data so they can create new text, images, code, summaries, and ideas in response to a prompt. Instead of only finding information, these tools help you **produce** it.

- **Common office uses:** drafting emails, rewriting reports, summarizing meetings, brainstorming ideas, and creating slide outlines
- **Key models:** Google Gemini, ChatGPT, and Claude — each offers different strengths for writing, reasoning, and productivity
- **Why it matters:** tasks that used to take hours can often be reduced to minutes with the right prompt
- **Practical example:** ask AI to turn a rough meeting note into a polished follow-up email or to summarize a 20-page document into 5 bullet points

The best results come when you give the model context, a clear goal, and a preferred format. For example: *"Summarize this spreadsheet for a manager in 3 bullets, using simple language."*

## How AI Models Are Trained

### Data Collection

Models learn from billions of text documents, code samples, and images. The broader the data, the more patterns the model can recognize.

### Pattern Recognition

The model identifies relationships between words, ideas, and structures so it can predict likely next steps in a response.

### Fine-Tuning

Training is adjusted for specific tasks such as writing, analysis, or coding so the model becomes more useful in real workflows.

### Continuous Improvement

New versions improve from feedback, fresh data, better instructions, and stronger safety guardrails.

### How this shows up at work

- Rewrite an email in a more professional tone
- Generate a spreadsheet formula or explain one in plain English
- Turn raw notes into a clean summary for a manager
- Create a first draft of a presentation agenda or project update

### Free practice data sources

Use free datasets to practice prompts, summaries, and analysis:

- **Kaggle Datasets**
- **Google Dataset Search**
- **data.world**

HOUR 1 — TOOLS DEEP DIVE

# Google Gemini & AI Knowledge Cutoffs

## Gemini for Workspace

Gemini is built directly into Google Workspace, so you can draft, analyze, summarize, and organize without leaving the apps people already use every day.

- **Works inside familiar tools:** Gmail, Docs, Sheets, Slides, and Drive
- **Practical examples:** Draft an email reply in Gmail, summarize meeting notes in Docs, create a slide outline in Slides, or analyze spreadsheet trends in Sheets
- **Time saved:** Turns repetitive writing and formatting tasks into a quick prompt-and-review workflow
- **Key features:** Help me write, Help me design, Help me organize
- **Gemini Business:** $20/user/month — **Enterprise:** $30/user/month
- **Large context window:** Gemini 1.5 Pro supports up to **1 million tokens**, which helps with long documents, transcripts, and complex reference material

## Knowledge Cutoffs — What AI Doesn't Know

A knowledge cutoff is the point after which a model may not reliably know new facts, product changes, or current events. That is why recent claims should always be checked against a trusted source.

- **Gemini 1.5 Pro:** Trained through early 2024
- **ChatGPT-4:** Cutoff April 2024; can browse the web for current information when enabled
- **Claude 3:** Early 2024; especially strong for long-form analysis and reasoning
- **Best practice:** Use AI for first drafts, then verify dates, pricing, policies, and any fast-changing facts
- **Example:** Ask AI to summarize a policy change, then confirm the final details on the official website before sharing it
- **Example:** For market research, use AI to structure the analysis, but validate numbers with a live dataset or recent report

Training tip: For practice, use free data sources like Kaggle or Google Dataset Search to test prompts and verify answers against real datasets.

Source links for practice: **Kaggle Datasets** and **Google Dataset Search**.

HOUR 1 — WORKFLOW DEMO

# ChatGPT for Office Work: Email Drafting Step-by-Step

Use ChatGPT to turn a rough complaint, request, or internal note into a polished professional email in minutes — with better tone, clearer structure, and less back-and-forth.

### 1. Paste the source text

Start with the messy input: a customer complaint, an internal note, a chat message, or a rough outline.

- Keep the original wording so the model can preserve facts
- Include the audience if you know it: customer, manager, vendor, or teammate
- Example: “Draft a reply to this upset customer about a delayed order.”

### 2. Add clear instructions

Tell ChatGPT the tone, length, and purpose so the draft matches the situation.

- Tone: calm, apologetic, firm, friendly, or executive
- Length: short reply, detailed explanation, or bullet summary
- Goal: reassure, request approval, explain a delay, or confirm next steps

### 3. Review 3–4 options

Ask for multiple versions when the message is sensitive or important.

- Choose the one that best fits the relationship and urgency
- Blend parts from different versions if needed
- Example: one version can be warmer, another more concise, another more formal

### 4. Customize and send

Check names, dates, promises, and policy details before sending.

- Replace placeholders with real facts
- Make sure the action request is specific
- Read aloud once to catch awkward phrasing

Professional response in **2 minutes** instead of 15 minutes of writing from scratch. The same workflow works for meeting prep, vendor comparisons, and product announcements.

## Practical prompt examples

- **Email drafting:** “Draft a professional response to this client complaint. Keep it calm, acknowledge the issue, and offer next steps.”
- **Meeting prep:** “Summarize these 5 emails into 3 talking points for tomorrow’s call.”
- **Content creation:** “Write a product announcement for our new feature in a clear, upbeat tone.”
- **Research:** “Compare these three vendors and give a recommendation with pros, cons, and risks.”

## Best practices for training and practice

- Use realistic examples from work: complaints, approvals, scheduling, or status updates
- Test the same prompt with different tones to see how output changes
- Have learners rewrite the prompt once to improve clarity and specificity
- Always verify facts, commitments, and deadlines before sending

Free dataset sources for practice: **Kaggle Datasets** and **Google Dataset Search**.

## Example workflow in one prompt

“Use the complaint below, draft 3 email reply options for a customer support manager, keep each under 120 words, and make one version more empathetic, one more formal, and one more concise.”

HOUR 2

# AI in Spreadsheets

Use AI to turn spreadsheet work from repetitive setup into a faster, more reliable workflow. Instead of building everything by hand, you can ask Gemini in Sheets to create structure, formulas, formatting, and summaries for you.

### Before AI

Typical spreadsheet work often means:

- Enter or clean hundreds of rows manually
- Build formulas one cell at a time
- Apply formatting and conditional rules by hand
- Check totals and fix formula mistakes

## 2 hrs

of tedious manual work

Example: creating a weekly expense tracker from scratch, then copying formulas across multiple tabs.

### With Gemini in Sheets

Prompt: *"Create a sales tracker with columns for date, product, amount, region, and calculate totals by region."*

- Create a full table layout automatically
- Generate SUM, IF, and other formulas correctly
- Suggest pivot tables and charts for analysis
- Apply conditional formatting to highlight trends

## 2 min

Full table, SUM formulas, pivot table, and conditional formatting — done.

Gemini in Sheets can create new sheets, complete end-to-end tasks on existing sheets, and generate precise formulas automatically.

Practical examples:

- Build a budget tracker with category totals
- Summarize customer feedback by sentiment or region
- Prepare a project status sheet with overdue items highlighted

For practice data, try free datasets from Kaggle or Google Dataset Search: **Kaggle Datasets** and **Google Dataset Search**.

HOUR 2 — TOOLS

# Gemini & ChatGPT in Your Spreadsheets

Use AI to move faster in Sheets and Excel — from building clean tables to generating formulas, spotting patterns, and preparing analysis-ready data.

## Gemini in Google Sheets

Gemini is strongest when you want to **create, transform, or analyze** a sheet directly inside Google Sheets.

### → Create Tables

Prompt example: “Build a project tracker with task, assignee, deadline, status, and priority.”

Best for quickly setting up a structure before you start entering data.

### → Generate Formulas

Prompt example: “Create a formula to calculate ROI for each campaign and explain how it works.”

Useful when you need formulas for totals, percentages, lookups, or automated calculations.

### → Format Data

Prompt example: “Apply conditional formatting to highlight sales above $50K and flag overdue tasks in red.”

Helps make the sheet easier to read, scan, and present to others.

### → Analyze Trends

Prompt example: “Run a full analysis on this sales dataset and identify trends by month and region.”

Great for surfacing outliers, recurring patterns, and early warnings.

### Practical examples

- Turn a messy CSV into a clean tracker with headers, formulas, and filters.
- Ask Gemini to build a monthly budget sheet with category totals and warning flags.
- Generate a sales pipeline with stage-by-stage conversion formulas.
- Summarize a long column of notes into themes or next steps.

## ChatGPT for Excel & Sheets

ChatGPT is especially helpful for **reasoning, explanation, and multi-step workflows** before or after the data is in the sheet.

- **Multi-tab analysis:** Ask questions across entire workbooks, not just one sheet.
- **Formula explanation:** “Explain what this formula does and how to modify it for another column.”
- **Data cleaning:** “Find duplicates, normalize names, and flag missing values.”
- **Model building:** “Create a 3-statement financial model from this data.”
- **Scenario planning:** “Show best-case, base-case, and worst-case forecasts for the next quarter.”
- **Documentation:** “Write a short README explaining how this workbook works.”

### Ready-made use cases

- Sales dashboard with revenue, pipeline, and forecasts
- Budget tracker with automated overspending flags
- Customer database with follow-up reminders
- Project timeline with critical path and resource allocation
- Content calendar with status tracking and publishing dates

### Source links for free practice data

Kaggle Datasets · Google Dataset Search · Data.world Open Data

HOUR 3

# Data Visualization with AI

**60,000x**

Faster Processing

People understand visual patterns far faster than dense text tables.

**65%**

Visual Learners

Charts, dashboards, and maps make insights easier to remember and explain.

**3x**

Faster Decisions

Clear visuals help teams spot trends, outliers, and next steps more quickly.

**10 min**

Dashboard Time

AI can turn raw data into a working dashboard in minutes instead of hours.

## What AI can create for you

- Trend charts for revenue, traffic, usage, or performance over time
- Comparison charts for top products, categories, regions, or teams
- Breakdown charts for budget, funnel stages, or customer segments
- Interactive dashboards with filters for date range, region, and channel

## Practical prompt examples

Use prompts that name the chart type, the metric, and the audience.

- **Sales review:** "Create a line chart showing monthly revenue by region for the last 12 months."
- **Marketing analysis:** "Build a dashboard with conversion rate, cost per acquisition, and campaign spend."
- **Product insights:** "Show the top 5 features by adoption, with a filter for plan type."
- **Operations report:** "Visualize weekly support tickets, average response time, and backlog size."

## How to get better results

1. Start with a clear question, not just a dataset.
2. Ask for the chart type that fits the story you want to tell.
3. Specify dimensions like date, region, product, or customer segment.
4. Request formatting details such as labels, colors, and annotations.
5. Iterate: ask AI to simplify, highlight anomalies, or add a comparison view.

## Example workflow

A good workflow is: upload data, ask for a chart, review the visual, then refine the prompt.

- Upload a CSV of sales or customer data.
- Ask for a dashboard that summarizes the key metrics.
- Request specific views like "top 10 products" or "monthly trend."
- Use the output as a starting point for a presentation or report.

## Free dataset sources for practice

- **Kaggle Datasets** — large library of public datasets for business, finance, health, and more.
- **Google Dataset Search** — search engine for publicly available datasets across the web.
- **data.world** — community datasets and public data projects.

AI is most useful when you combine a clear question, a clean dataset, and one specific chart request.

*AI prompt example:* "Create a dashboard showing revenue trend, top 5 products, customer acquisition cost, and conversion rate" — AI delivers 4 color-coded charts with date range and region filters, in minutes.

HOUR 3 — CHART GUIDE

# AI Chart Tools & When to Use Each Chart Type

## AI Chart Generation Tools

**Gemini in Sheets:** Great for quickly turning spreadsheet data into a visual dashboard. Use a prompt like *"Create a visual dashboard of this sales data"* to generate summary charts, trends, and highlights automatically.

**ChartGen AI:** Best when you want to upload a dataset, describe the goal, and get polished charts in seconds. Helpful for quick drafts, client-ready visuals, and repeated chart creation.

**PlotSet:** Useful when you need a larger library of chart templates. Its AI text-to-chart workflow helps you move from raw data to a finished chart with less manual setup.

## Building Interactive Dashboards

- Combine revenue, pipeline, and acquisition metrics on one page for a complete view
- Add filters so viewers can drill down by region, product, campaign, or time period
- Connect to live data sources so charts refresh automatically as the data changes
- Export as PNG or PDF, or embed dashboards directly into presentations and reports
- Keep one chart per question: ask what decision the audience needs to make before choosing the visual

## When to Use Each Chart Type

**Bar charts:** Best for comparing categories. Example: sales by region, support tickets by team, or top-performing products.

**Line charts:** Best for trends over time. Example: monthly revenue, website traffic, churn rate, or daily app usage.

**Pie charts:** Best for proportions of a whole when there are only a few slices. Example: market share, budget allocation, or channel mix.

**Scatter plots:** Best for showing relationships or correlations. Example: price vs demand, experience vs salary, or ad spend vs conversions.

**Practical rule:** If the audience needs to compare, use bars. If they need to track change, use lines. If they need composition, use pie or stacked charts. If they need to understand relationships, use scatter plots.

### Training Practice

Try a free dataset from **Kaggle Datasets** or **Google Dataset Search**, then build the same question three ways: one trend chart, one comparison chart, and one relationship chart.

HOUR 4

# Hands-On Practice: Three Exercises

Use these exercises to practice prompting, structuring, and iterating with AI tools. Each one mirrors a real work task, so focus on clarity, context, and the quality of the final output.

1

### Draft an Email

**Scenario:** Respond to a difficult customer complaint.

**Task:** Ask Gemini or ChatGPT to draft a professional, empathetic response that acknowledges the issue, takes responsibility where appropriate, and offers a clear next step.

- Include the customer's concern, tone, and the desired outcome
- Ask for 2 versions: one concise and one more formal
- Example prompt: "Write a calm, professional reply to a customer who received the wrong order and is asking for a refund."

**Time:** 10 minutes → **Outcome:** A polished email you could send with only minor edits.

2

### Build a Spreadsheet

**Scenario:** Create a quarterly sales tracker with formulas and analysis.

**Task:** Use Gemini in Sheets to build the structure, add calculations, and identify trends that matter to your team.

- Start with columns for month, rep, region, revenue, and target
- Ask for formulas like totals, averages, and percentage-to-goal
- Example prompt: "Create a sales tracker with monthly totals, quarterly rollups, and a column that flags deals below target."

**Time:** 15 minutes → **Outcome:** An automated sheet that updates when new data is added.

3

### Create a Dashboard

**Scenario:** Visualize your team's performance metrics.

**Task:** Use AI chart tools to build 3–4 charts, then combine them into a dashboard that tells a clear story.

- Choose metrics with a purpose: trend, comparison, composition, or correlation
- Use consistent labels, colors, and chart types for readability
- Example prompt: "Turn this dataset into a stakeholder dashboard showing revenue trend, conversion rate, top regions, and monthly growth."

**Time:** 20 minutes → **Outcome:** A professional dashboard ready for stakeholders.

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### Free practice datasets

Use a public dataset if you want real data to practice with:

- **Kaggle Datasets** — searchable library of free datasets across business, finance, marketing, and operations
- **Google Dataset Search** — find open datasets from universities, government sources, and organizations
- **data.world Free Datasets** — browse public datasets for analysis and visualization practice

Tip: When practicing, first describe the audience, then the goal, then the constraints. Better prompts usually lead to better outputs.

HOUR 4 — SKILL BUILDER

# Prompt Engineering: The Art of Asking

## Prompt Refinement in Action

Great prompts evolve from broad requests into clear instructions that define the task, the format, and the expected outcome.

1

### Round 1 — Too Vague

"Analyze this sales data"

- No context about what matters
- No target audience or decision goal
- Leaves the model guessing what "analyze" means

2

### Round 2 — Better

"Analyze sales data to find top-performing regions"

- Adds a clear focus area
- Gives the model a measurable objective
- Still leaves out time period, output format, and depth

3

### Round 3 — Excellent

"Analyze Q1–Q4 sales by region. Identify top 3 by revenue, calculate growth rate, and flag declining regions."

- Specifies the time range and metric
- Defines the exact output needed
- Helps produce a response that is ready to use in a report or meeting

## The Prompt Formula

Use a simple structure to make prompts more reliable, reusable, and useful across tools like Gemini, ChatGPT, and Copilot.

### ❌ Vague

"Make a chart"

- Too broad to produce a strong result
- Doesn't say what data to use
- Doesn't define chart type, audience, or style

### ✅ Good

"Create a bar chart comparing Q1–Q4 revenue by product category, sorted highest to lowest"

- Names the chart type
- Defines the data and comparison
- Includes an ordering rule for clarity

### ⭐ Great

"Create a bar chart of Q1–Q4 revenue by category, sorted descending, with a growth trend line. Highlight products above $100K in green."

- Adds styling and emphasis rules
- Combines chart type with interpretation
- Produces a more presentation-ready output

**Key lesson:** More specific detail = more useful results. Always include context, format preferences, and desired output.

### Practical Prompt Template

Try this pattern when writing prompts for work:

1. **Role or context:** "You are an analyst helping a sales manager…"
2. **Task:** "Summarize Q2 customer feedback…"
3. **Constraints:** "Use 3 bullets and avoid jargon…"
4. **Output format:** "Return a table with insights, evidence, and next steps."

### Free Practice Data Sources

Use public datasets to practice prompt writing, analysis, and chart creation:

- **Kaggle Datasets** — large library of free datasets across business, health, education, and more
- **Google Dataset Search** — search engine for finding public datasets from trusted sources
- **UCI Machine Learning Repository** — classic open datasets for experimentation and practice

# Best Practices & Common Pitfalls

## Best Practices

### Be Specific

State the task, the audience, the format, and the success criteria. For example: "Create a bar chart comparing Q1–Q4 revenue by product category, sorted highest to lowest, with a short summary of the biggest trend."

- Include the topic, time period, and output type
- Specify any constraints, like tone, length, or style
- Mention what to highlight, compare, or exclude

### Iterate

Treat the first answer as a draft. Follow up with targeted requests to sharpen the result, add missing context, or change the level of detail.

- "Make it shorter and more executive-friendly"
- "Add a comparison against last quarter"
- "Turn this into a step-by-step checklist"

### Combine Tools

Use the right tool for each stage of the workflow. One tool can draft, another can analyze, and a third can help with final formatting and polish.

- Use AI writing tools for drafting and brainstorming
- Use analysis tools for summarizing data or spotting patterns
- Use presentation tools to refine layout, visuals, and messaging

## Pitfalls to Avoid

### Over-Reliance

AI can accelerate work, but it should not replace your judgment. Always ask whether the output makes sense for your audience and business goal.

- Use AI as a starting point, not the final authority
- Review recommendations before acting on them
- Watch for confident-sounding but incorrect answers

### Data Privacy

Never paste sensitive customer, employee, legal, or financial data into public AI tools unless it has been approved and properly anonymized.

- Remove names, account numbers, and identifiers
- Use sample or synthetic data when possible
- Follow your organization’s security policy

### Always Verify

Check formulas, citations, calculations, and generated claims before you share them. Small errors can turn into big problems if they reach a client or leadership team.

- Cross-check facts against trusted sources
- Test formulas with a small example
- Read outputs for missing assumptions or edge cases

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Free dataset sources for practice: **Kaggle Datasets** and **Google Dataset Search**.